Sanaul Haque

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A Baseline-Normalized Satellite Framework for Early Detection and Forecasting of Industrial River Pollution

Full Name: Ayva Patel

Institution: Avenues: The World School

Category: Junior

Country: United States of America

Email: ayva.patel.uk@gmail.com

Research Topic: Environmental Science/Remote Sensing

Research Problem

Industrial textile production can substantially degrade surrounding waterways through wastewater containing suspended solids, organic matter, dyes, and surfactants. Conventional water-quality monitoring relies primarily on in-situ sampling and laboratory analysis. Although accurate, these methods are costly, labor-intensive, and spatially limited, making continuous monitoring difficult across large industrial regions [1].

Satellite remote sensing provides a scalable alternative for environmental screening. Sentinel-2 offers repeated multispectral observations at spatial resolutions as fine as 10 m and has been used to monitor optically active water characteristics such as turbidity and suspended sediment [3], [5], [6]. However, two major problems limit its use for industrial pollution monitoring. First, rivers naturally differ in sediment load, hydrology, seasonality, and optical characteristics, making universal spectral thresholds unreliable. Second, chemical parameters such as chemical oxygen demand (COD), biochemical oxygen demand, and surfactant concentration cannot be directly quantified from multispectral reflectance without local in-situ calibration [1], [4].

This research therefore investigates a different approach: rather than estimating exact pollutant concentrations, can satellite observations identify when downstream conditions become statistically abnormal relative to both upstream conditions and a river's own historical behavior?

The proposed framework combines upstream–downstream comparison, historical baseline normalization, statistical anomaly detection, and trend forecasting. Its purpose is not to replace laboratory monitoring, but to provide a scalable early-warning system that can identify where and when targeted field investigation may be necessary.

Existing Literature and Research Gap

Satellite platforms including Landsat, MODIS, and Sentinel-2 have increasingly been applied to water-quality monitoring [2], [7]. Turbidity is particularly suitable for remote sensing because suspended material alters visible reflectance. The Normalized Difference Turbidity Index (NDTI), which compares red and green reflectance, has therefore been used to assess turbidity and suspended sediment [5], [6].

Industrial wastewater presents a more difficult problem. COD, BOD, and surfactant concentrations are not directly optically measurable, and relationships between reflectance and these parameters may vary among rivers [1], [4]. Meanwhile, large-scale satellite monitoring and anomaly analysis have advanced [7], and early-warning systems have demonstrated the value of identifying deteriorating environmental conditions before severe impacts occur [8].

However, integration of satellite observations, simultaneous upstream referencing, historical normalization, statistical validation, and predictive trend modeling within a single industrial-river monitoring framework remains limited. This research addresses that gap by allowing each river to serve partly as its own control. Rather than applying the same absolute spectral threshold

everywhere, downstream–upstream differences will be compared with their own historical distributions.

Research Question and Hypotheses

The central research question is:

Can an upstream-referenced, baseline-normalized satellite framework reliably detect and forecast abnormal downstream water-quality deterioration associated with textile-industrial activity across heterogeneous river systems?

Three supporting questions will be investigated: (1) Are satellite-derived water-quality indicators significantly elevated downstream of textile-industrial areas relative to upstream reference locations? (2) Can historical normalization distinguish persistent spatial differences from disproportionate downstream deterioration? (3) Can standardized anomaly trajectories identify and forecast warning-level conditions?

Three hypotheses will be tested:

- H1: Pollution-associated satellite indicators will be significantly greater downstream than upstream.
- H2: For indicators experiencing localized deterioration, downstream conditions will worsen more strongly over time than corresponding upstream conditions.
- H3: Normalizing downstream–upstream differences against site-specific historical distributions will enable standardized anomaly classification and time-to-warning estimation without requiring universal spectral thresholds.

Methodology

Study Design and Satellite Data
A longitudinal, multi-region analysis will examine textile-intensive river systems in Bangladesh, Indonesia, and China. Using geographically and hydrologically distinct regions will test whether the framework can function beyond a single watershed.

Each region will contain an upstream reference zone and a downstream zone approximately 1–7 km from textile-industrial clusters. Major municipal discharge points, wastewater-treatment facilities, and river confluences will be avoided where possible. River segments wider than approximately 30 m will be prioritized to reduce mixed land-water pixels.

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Sentinel-2 Level-2A surface-reflectance imagery from 2017–2025 will be analyzed using B2 (blue, 490 nm), B3 (green, 560 nm), and B4 (red, 665 nm). Scenes with more than 50% cloud cover and same-day duplicates will be excluded. River polygons will isolate water pixels before reflectance values are spatially averaged. The dataset contains 351 independent dates for

Bangladesh, 283 for Indonesia, and 703 for China, providing a large temporal dataset across three regions.

Three spectral indicators will be evaluated:

NDTI = $(B4 - B3) \div (B4 + B3)$ for turbidity [5];

$R_{organic}$ = B3 ÷ B2 as an exploratory organic-load-associated optical proxy; and

R_{foam} = B3 ÷ B4 as an exploratory surface-foam/surfactant-associated optical proxy.

The latter two will not be interpreted as direct chemical measurements. Instead, they will be treated as optical anomaly indicators requiring future in-situ calibration.

Statistical Analysis and Early-Warning Framework
First, one-tailed independent two-sample t-tests will determine whether each indicator is significantly elevated downstream:

H_0: $\mu D \leq \mu U$
H_1: $\mu D > \mu U$

2 Statistical significance will be evaluated at $\alpha = 0.05$, accompanied by 95% confidence intervals and Cohen's d effect sizes.

However, a higher downstream value alone does not establish recent deterioration. A river may naturally have persistent upstream–downstream differences. Temporal change will therefore also be evaluated using:

$\Delta\Delta I = (D_{current} - D_{baseline}) - (U_{current} - U_{baseline})$

A positive ΔΔI indicates that downstream conditions increased more strongly than upstream conditions, helping distinguish localized deterioration from broader environmental change.

For early-warning detection, a differential impact signal will be calculated for each observation:

$\Delta I(t) = Id(t) - Iu(t)$

The 2017–2018 period will establish the historical mean ($\mu\Delta,0$) and standard deviation ($\sigma\Delta,0$). Each subsequent observation will then be standardized:

$Z(t) = [\Delta I(t) - \mu\Delta,0] \div \sigma\Delta, 0$

A three-observation moving average will reduce sensitivity to isolated fluctuations. Conditions will then be classified as:

Normal: $Z < 1$

Watch: $1 \leq Z < 2$
Warning: $Z \geq 2$

Unlike a universal spectral threshold, these classifications identify when a river's downstream impact becomes abnormal relative to its own historical behavior.

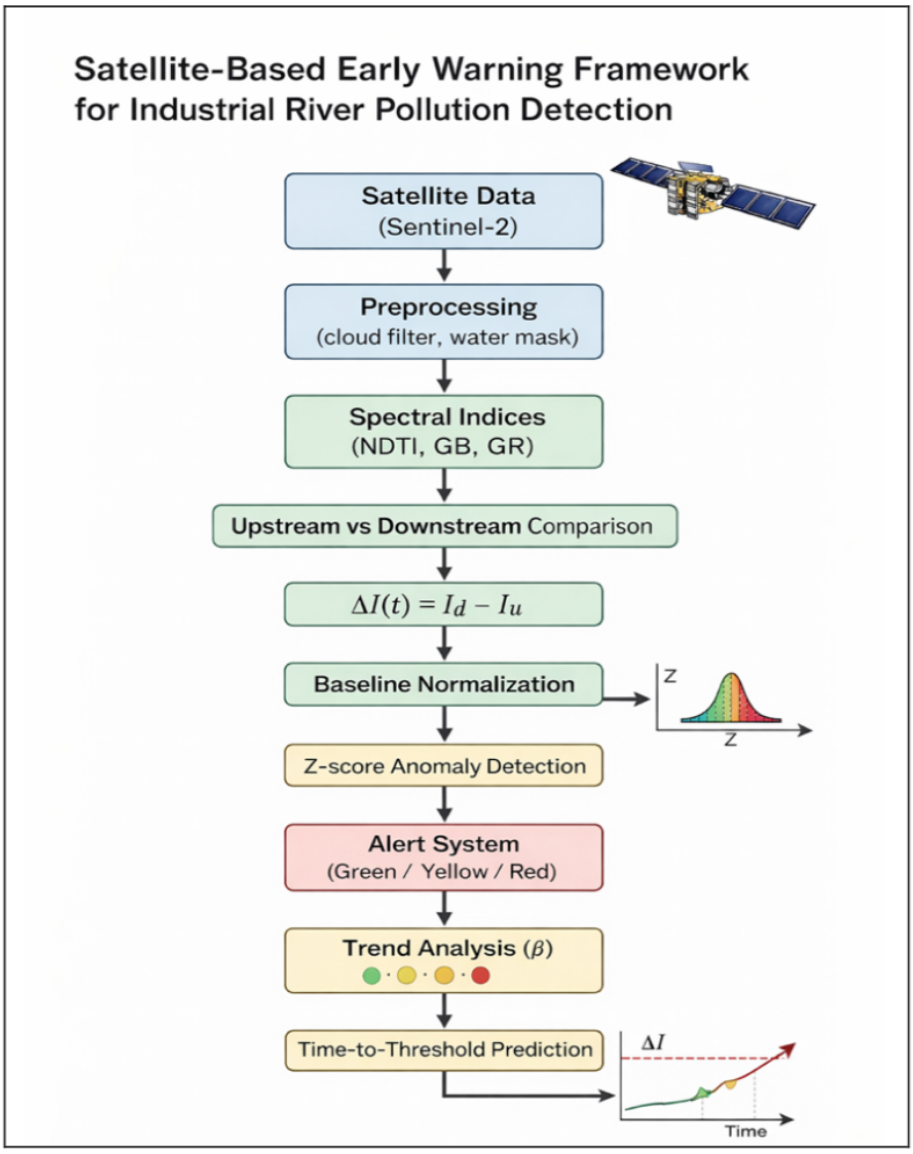


Figure 1. Proposed baseline-normalized satellite early-warning framework.

Following anomaly detection, a temporal regression will evaluate whether standardized anomalies are increasing:

$Z(t) = \beta_0 + \beta_1 t + \varepsilon$ with H_0: $\beta_1 \leq 0$ and H_1: $\beta_1 > 0$.

For positive trends, the current anomaly and estimated slope will be used to calculate the projected time to a Watch or Warning threshold. Confidence intervals around β_1 will provide an uncertainty range rather than a single deterministic prediction. The framework therefore progresses from detection → normalization → classification → forecasting, as summarized in Fig. 1.

Expected Outcomes, Innovation, and Significance

The study is expected to demonstrate that downstream elevation and temporal deterioration are distinct phenomena. Some rivers may have consistently elevated downstream indicators without continued deterioration, whereas others may show an increasing downstream anomaly after corresponding upstream changes are controlled.

The central innovation is not a claim that satellites can directly measure industrial chemicals. Instead, it is the integration of upstream referencing, site-specific historical normalization, statistical validation, standardized anomaly detection, and time-to-threshold forecasting into one monitoring framework.

This design has three principal advantages. First, upstream referencing helps control for environmental changes affecting an entire river system. Second, baseline normalization avoids assuming that a single spectral pollution threshold applies across different rivers. Third, forecasting extends satellite analysis beyond retrospective mapping by identifying whether abnormal conditions are intensifying toward warning levels.

Several limitations remain. The organic-load and surfactant-associated indicators are exploratory optical proxies rather than direct chemical measurements. Rainfall, discharge, seasonality, cloud cover, atmospheric effects, and mixed pixels may influence satellite observations. Additionally, trend forecasts are conditional on recent trajectories and should not be interpreted as deterministic predictions. Future validation should pair satellite observations with simultaneous laboratory measurements and incorporate hydrological variables.

Nevertheless, because Sentinel-2 provides repeated global observations, the framework could eventually offer a comparatively low-cost complement to conventional monitoring. Rather than replacing laboratory analysis, it could help identify where and when limited field-monitoring resources should be deployed.

The broader objective is to shift satellite industrial-water monitoring from asking “What exact pollutant concentration is present?” toward a question better suited to remote sensing:

“Is this river becoming abnormally different from its historical condition, and is that abnormality accelerating?”

If successful across Bangladesh, Indonesia, and China, this approach could provide the foundation for a scalable satellite-based early-warning system for industrial river pollution.

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